

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2022-23

Paper: CS-103 — Numerical Computing

Time: Three hours

Maximum Marks: 70

Instructions: ATTEMPT ANY FIVE QUESTIONS. THE FIGURES IN THE RIGHT-HAND MARGIN INDICATE

Question 1.

- (a) Express the following decimal numbers in binary, octal and hexadecimal: [6]
(i) 22.623
(ii) -10.215
- (b) What are the three different types of errors encountered in numerical calculations? Explain each of them briefly. [3]
- (c) Determine the roots of the following equation using Newton-Raphson method correct to three decimal places: [5]

$$x^3 - 5x + 3 = 0$$

Question 2.

- (a) Derive the Regula-Falsi method for calculation of roots of non-linear equation. Write the steps of calculation of roots using this method. Discuss the scenario where this method would give large errors. [8]
- (b) Calculate the roots of the following equation using Regula-Falsi method, correct to three decimal places: [6]

$$x^3 + x^2 + x + 7 = 0$$

Question 3.

- (a) Describe the LU decomposition method for solving system of linear equations. [2]
- (b) Use the LU decomposition method to solve the following system of linear equations: [7]

$$\begin{aligned} 5x - 2y + z &= 4 \\ 7x + y - 5z &= 8 \\ 3x + 7y + 4z &= 10 \end{aligned}$$

- (c) Describe the method of diagonalization of a real symmetric matrix using Jacobi method. [5]

Question 4. For the matrix A given below, calculate:

$$A = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}$$

(a) The largest Eigen value and the corresponding Eigen vector. [7]

(b) Inverse of the matrix. [7]

Question 5.

(a) Derive Simpson's 1/3rd Rule for numerical integration. [4]

(b) Integrate the following function using:

$$I = \int_0^1 \frac{1}{1+x} dx$$

(i) Trapezoidal Rule

(ii) Simpson's 1/3rd Rule

using $h = 0.125$. Also compare the results obtained in each case with the actual value of the integral. [10]

Question 6.

(a) Derive the Newton-Gregory forward interpolation formula. [4]

(b) Use Newton's Divided difference method to calculate $f(0.25)$ and $f(0.27)$ for the following tabulated function: [8]

x	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

(c) State two applications of interpolation. [2]

Question 7.

(a) Use Picard's method to solve the differential equation:

$$\frac{dy}{dx} = 1 + xy, \quad y(0) = 1$$

Find the value of y for $x = 0.5$ correct to 3 decimal places. [7]

(b) Use Predictor-Corrector method to solve the following differential equation:

$$\frac{dy}{dx} = y(1 + x^2), \quad y(0) = 1$$

Use $h = 0.05$ and compute $y(0.3)$ correct to 3 decimal places. [7]